Software Testing — Complete Notes

Quality concepts · Unit, integration, E2E, TDD, BDD & more

Topic Testing | Goal Catch bugs early | Levels Unit→Integration→E2E | Approaches TDD / BDD

A reader-friendly, all-in-one reference for software testing concepts.

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1. 🎯 Why Testing

Software testing is the practice of **checking that code behaves as intended** — automatically and repeatably. Good tests catch bugs early (when they're cheap to fix), document expected behavior, enable confident refactoring, and are the safety net that makes **CI/CD** possible.

> 💡 **Mental model:** tests are an **executable specification**. They say "given this input, expect this output" and fail loudly when reality drifts. The earlier a bug is caught, the cheaper it is — a unit test catches it in seconds; production catches it in an incident.

Goals: correctness, prevent regressions, enable safe change, document behavior, and give fast feedback.

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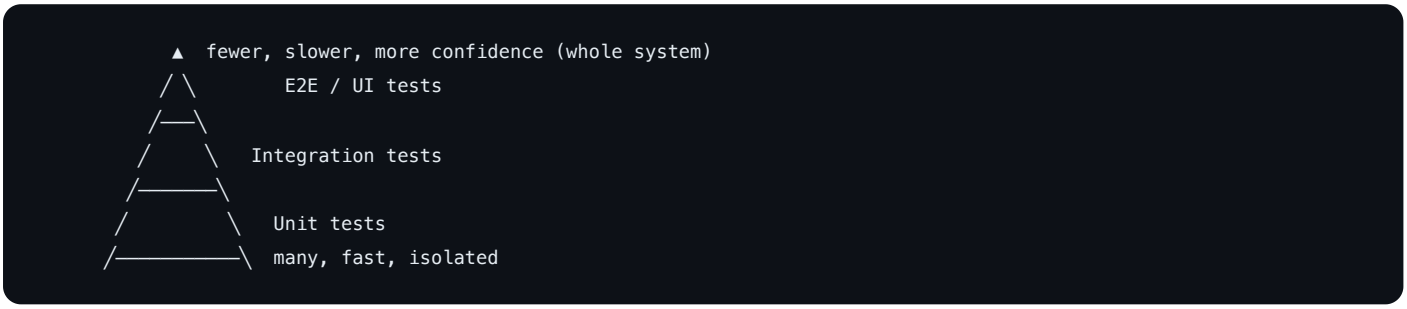
2. 🔺 The Testing Pyramid

A guideline for **how many** of each test type to write: many fast low-level tests, few slow high-level ones.



Level	Speed	Scope	How many
Unit	ms	One function/class in isolation	Most (broad base)
Integration	seconds	Modules together (DB, API)	Some (middle)
E2E	minutes	Whole app, user flows	Few (thin top)

> ⚠️ Anti-patterns: the **ice-cream cone** (too many slow E2E, few units) and the **hourglass** (lots of unit + E2E, no integration). Aim for a wide, fast base.

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3. 🧪 Unit Testing

- Tests **one small unit** (a function/method/class) **in isolation** from its dependencies.
- **Fast, deterministic, numerous** — run on every save/commit.
- External dependencies (DB, network, time) are replaced with **test doubles** (mocks/stubs).
- Structure: **Arrange** → **Act** → **Assert** (AAA), or Given-When-Then.

```
# Arrange-Act-Assert
def test_add():
    calc = Calculator()          # arrange
    result = calc.add(2, 3)      # act
    assert result == 5           # assert
```

> Good unit tests are **FIRST**: Fast, Isolated, Repeatable, Self-validating, Timely.

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4. 🔗 Integration Testing

- Verifies that **multiple components work together** — e.g. service ↔ database, API ↔ external service, module ↔ module.
- Catches bugs unit tests can't: wrong SQL, serialization mismatches, wiring/config errors, contract mismatches.
- Slower than unit tests (real DB/queue, often via **containers** like Testcontainers); fewer of them.
- May use a **real test database** or in-memory substitute.

> The line: a **unit test** mocks the database; an **integration test** uses a real (test) database to verify the queries actually work.

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5. 🌐 End-to-End (E2E) Testing

- Tests the **entire application from the user's perspective**, through the real UI/API, with real (or production-like) dependencies.
- Validates complete **user journeys** (sign up → add to cart → checkout).
- **Slowest, most brittle, most realistic** — keep few, cover critical paths only.
- Tools drive a real browser (Cypress, Playwright, Selenium).

> E2E gives the highest confidence but the worst feedback loop and maintenance cost — reserve it for the handful of flows that absolutely must work.

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6. 🧰 Other Test Types

Type	Checks
Smoke	Does the build basically work? (quick sanity before deeper tests)
Regression	Did a change break previously working features?
Acceptance (UAT)	Does it meet business/user requirements?
Performance / Load / Stress	Speed, throughput, behavior under heavy/extreme load
Security	Vulnerabilities (SAST/DAST, pen testing)
Smoke vs Sanity	Smoke = broad shallow build check; Sanity = narrow deep check of a specific fix
Exploratory	Unscripted human investigation
Accessibility (a11y)	Usable by people with disabilities
Snapshot	Output matches a saved reference (UI components)

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7. ⚖️ Functional vs Non-Functional

	Functional	Non-Functional
Asks	Does it do **what** it should?	**How well** does it do it?
Examples	Login works, cart totals correctly	Speed, scalability, security, usability, reliability
Types	Unit, integration, E2E, acceptance	Performance, load, stress, security, accessibility

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8. 🔴🟢 TDD (Test-Driven Development)

Write the **test first**, then the code to pass it. The cycle is **Red** → **Green** → **Refactor**:

1. **Red** — write a small failing test for the next bit of behavior.
2. **Green** — write the simplest code that makes it pass.
3. **Refactor** — improve the design while keeping tests green.

> **Benefits:** forces testable design, gives instant feedback, builds a safety net, prevents over-engineering (you only write code a test demands). **Cost:** discipline + slower upfront. TDD is a *design* technique as much as a testing one.

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9. 🥒 BDD (Behavior-Driven Development)

An evolution of TDD that describes behavior in **plain, business-readable language** so devs, QA, and product share one spec.

- Scenarios use **Given–When–Then** (Gherkin syntax):

```
Feature: Checkout
  Scenario: Successful purchase
    Given a customer with an item in the cart
    When they complete the checkout
    Then the order is confirmed
    And a receipt email is sent
```

- Tools (**Cucumber**, **SpecFlow**, **Behave**) map each Gherkin step to code.
- **TDD vs BDD:** TDD tests *units* in code terms ("function returns 5"). BDD tests *behavior* in business terms ("order is confirmed") — improves collaboration and living documentation.

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10. 🧭 Other Approaches (ATDD, etc.)

Approach	Idea
ATDD (Acceptance Test-Driven)	Define acceptance criteria as tests *before* building; whole team agrees on "done".
Contract testing	Verify the agreement between a service and its consumers (e.g. Pact) — catches breaking API changes.
Property-based testing	Assert properties over **many auto-generated inputs** (Hypothesis, QuickCheck) instead of fixed examples.
Mutation testing	Introduce bugs ("mutants") to check your tests actually catch them — measures test *quality*.
Fuzz testing	Feed random/malformed input to find crashes/security holes.
Snapshot testing	Compare output against a stored baseline.

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11. 🎭 Test Doubles (Mocks, Stubs…)

"Test double" = any stand-in for a real dependency. Types (by Gerard Meszaros):

Double	Purpose
Dummy	Passed but never used (fills a parameter).
Stub	Returns canned answers to calls.
Spy	A stub that also **records** how it was called.
Mock	Pre-programmed with **expectations**; the test *verifies* it was called correctly.
Fake	A working lightweight implementation (e.g. in-memory DB).

> Rule of thumb: **stub** to control inputs ("when called, return X"); **mock** to assert interactions ("expect this was called once with Y"). Don't over-mock — it couples tests to implementation.

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12. 📊 Coverage & Quality Metrics

- **Code coverage** = % of code exercised by tests (line, branch, statement, function).
- High coverage ≠ good tests — it shows what ran, **not** that assertions are meaningful. 100% coverage with weak asserts is false confidence.
- **Mutation score** is a better quality signal (do tests catch injected bugs?).
- Aim for **meaningful coverage of critical paths**, not a vanity number; watch **flaky tests** (pass/fail nondeterministically) — they erode trust and must be fixed or quarantined.

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13. 🔄 Testing in CI/CD

- Tests are the **gate** in a pipeline — a merge/deploy only proceeds if they pass. See the CI/CD notes.
- **Fail fast:** run cheap, broad tests first (lint → unit), slow/narrow ones later (integration → E2E).
- Run unit tests on **every push/PR**; integration in CI with **ephemeral containers**; E2E on a deployed staging environment.
- **Shift left** — test as early as possible; publish results + coverage as build artifacts; block merges on red.

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14. 🛠️ Tools by Language

Language / area	Common tools
JavaScript/TS	Jest, Vitest, Mocha, Playwright, Cypress, Testing Library
Python	pytest, unittest, Hypothesis, tox
Java	JUnit, TestNG, Mockito, REST Assured
Go	built-in `testing`, testify
C#/.NET	xUnit, NUnit, MSTest, Moq
BDD	Cucumber, SpecFlow, Behave
E2E/Browser	Cypress, Playwright, Selenium
Load	k6, JMeter, Gatling, Locust

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15. ✅ Best Practices

- Follow the **pyramid** — many fast unit tests, fewer integration, few E2E.
- One **clear assertion focus** per test; name tests by behavior ("returns_error_when_empty").
- Keep tests **FIRST** (Fast, Isolated, Repeatable, Self-validating, Timely) and **deterministic** (no real time/network/random).
- **Arrange-Act-Assert** structure; avoid logic in tests.
- Test **behavior, not implementation** — don't over-mock; tests shouldn't break on every refactor.
- Fix or quarantine **flaky tests** immediately.
- Run tests in **CI on every PR**; fail fast; track meaningful coverage of critical paths.
- Treat **test code as production code** — review it, refactor it, keep it clean.

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16. 🧠 Quick Mental Model

- **Tests = executable spec; catch bugs early, enable safe change.**
- **Pyramid:** lots of **unit** (fast, isolated) → some **integration** (components together) → few **E2E** (whole app, user flows).
- **Unit mocks the DB; integration uses a real test DB; E2E drives the real UI.**

- **TDD** = Red→Green→Refactor (test first, design tool). **BDD** = Given-When-Then in business language (collaboration + living docs).
- **Test doubles:** stub (canned returns) vs mock (verify interactions); fake = lightweight real impl.
- **Coverage shows what ran, not test quality** — aim for meaningful asserts; kill flaky tests.
- Tests are the **gate in CI/CD** — fail fast, shift left, block merges on red.

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17. ❓ Common Interview Questions

Rapid-fire questions interviewers ask about testing:

- **Q: Unit vs integration vs E2E?** — Unit tests one isolated unit (mocked deps). Integration tests components together (real DB/API). E2E tests the whole app through the UI as a user.
- **Q: What is the testing pyramid?** — A guideline: many fast unit tests, fewer integration, few slow E2E — for fast feedback and stability.
- **Q: TDD cycle?** — Red (write a failing test) → Green (minimal code to pass) → Refactor (clean up, stay green).
- **Q: TDD vs BDD?** — TDD tests units in code terms; BDD describes behavior in business language (Given-When-Then) for shared understanding.
- **Q: Mock vs stub?** — A stub returns canned data (controls inputs); a mock has expectations you verify (asserts interactions).
- **Q: Does 100% code coverage mean good tests?** — No — coverage shows what code ran, not that assertions are meaningful. Mutation testing better measures quality.
- **Q: What's a flaky test?** — One that passes/fails nondeterministically; it erodes trust and must be fixed or quarantined.
- **Q: Functional vs non-functional testing?** — Functional = does it do the right thing; non-functional = how well (performance, security, usability).
- **Q: What is regression testing?** — Re-checking existing functionality still works after a change.
- **Q: Smoke vs sanity testing?** — Smoke = quick broad check the build works; sanity = narrow deep check of a specific fix/area.
- **Q: Where do tests fit in CI/CD?** — They're the gate: run on every PR, fail fast (unit→integration→E2E), block merges/deploys on failure.
